

HORSESHOE BAY, TX, USA

WMO: 720639

Lat: 30.533N

Lon: 98.367W

Elev: 1093

StdP: 14.12

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00225

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	28.4	11.9	10.5	38.0	16.4	13.1	39.8	19.5	55.1	17.1	53.7	4.7	0	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.4	100.7	71.4	99.3	71.7	97.3	71.9	76.5	87.0	75.8	86.5	75.2	86.0	6.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	134.1	78.2	73.3	128.8	78.6	72.6	126.0	78.5	40.6	87.7	40.0	86.6	39.4	84.9	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	14.6	12.5	19.3	104.1	5.1	3.1	15.6	106.4	12.7	108.2	9.8	110.0	6.1	112.2		
(5)			16.9	78.1	4.0	1.1	14.0	78.9	11.7	79.6	9.5	80.3	6.6	81.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.4	49.4	53.9	61.6	68.5	75.0	82.3	84.8	85.6	79.2	69.4	58.8	51.1		
	DBStd	14.80	9.33	11.05	9.13	7.27	6.65	3.60	3.17	3.70	5.41	8.31	9.68	9.27		
	HDD50	354	124	79	17	0	0	0	0	0	0	2	32	100		
	HDD65	1774	489	331	172	47	11	0	0	0	1	53	230	440		
	CDD50	7065	105	189	378	556	776	969	1079	1103	875	604	297	134		
	CDD65	3008	4	21	67	154	322	519	614	638	426	190	44	9		
	CDH74	34595	66	172	563	1384	3068	6137	7952	8523	4556	1807	287	80		
	CDH80	17547	6	35	129	441	1284	3175	4510	4985	2270	647	57	8		
(14) Wind	WSAvg	5.2	4.7	5.4	6.0	6.5	6.3	6.0	5.2	4.8	4.0	4.6	4.6	4.3		
(15) Precipitation	PrecAvg	30.6	1.8	1.8	2.6	2.5	3.8	3.2	2.1	1.9	2.8	3.2	2.6	1.8		
	PrecMax	47.1	4.6	6.4	8.3	6.5	7.8	11.7	9.5	9.9	7.3	10.5	9.9	9.8		
	PrecMin	13.1	0.0	0.0	0.0	0.3	0.8	0.6	0.0	0.0	0.1	0.0	0.0	0.0		
	PrecStd	8.6	1.3	1.7	1.7	1.7	1.9	2.6	2.1	2.1	1.9	2.7	2.3	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.9	84.7	87.3	91.7	97.1	101.8	103.6	104.2	100.1	93.0	86.2	81.3		
		MCWB	59.2	60.3	63.9	67.6	72.3	71.5	70.5	71.4	68.7	69.2	68.3	62.1		
	2%	DB	75.0	79.1	83.6	88.3	92.9	98.6	100.3	101.5	97.0	89.9	81.0	75.8		
		MCWB	58.5	60.6	64.7	67.8	71.1	71.8	71.2	72.0	70.3	71.1	65.6	62.5		
	5%	DB	70.5	74.9	80.6	85.1	90.2	95.4	98.8	99.5	94.1	86.7	77.0	71.8		
		MCWB	57.3	61.3	62.9	66.7	70.7	72.2	71.6	72.2	70.6	68.8	64.3	61.3		
	10%	DB	66.1	71.2	76.7	81.8	87.6	93.1	96.7	97.7	91.3	83.7	73.6	67.7		
		MCWB	55.1	60.9	61.9	65.3	70.8	72.6	72.1	72.2	71.0	67.1	64.2	60.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.9	68.8	69.3	74.1	76.4	77.6	77.3	76.9	77.0	75.9	72.7	69.1		
		MCDB	70.8	75.3	78.2	83.2	85.7	88.5	86.7	89.9	85.9	83.4	79.6	74.7		
	2%	WB	63.6	67.1	67.9	72.0	75.0	76.3	76.3	75.8	75.6	74.3	70.7	66.8		
		MCDB	69.6	72.4	76.0	81.5	84.3	87.2	86.4	88.6	85.1	82.0	76.1	71.5		
	5%	WB	60.4	65.1	66.7	70.5	73.9	75.4	75.4	75.2	74.9	72.9	68.4	64.0		
		MCDB	66.5	70.2	74.5	79.7	83.5	86.4	86.1	88.0	84.2	80.8	73.5	68.7		
	10%	WB	57.1	62.8	65.5	69.0	72.9	74.6	74.8	74.5	74.1	71.4	66.2	60.7		
		MCDB	65.0	68.4	73.4	77.6	82.6	85.8	85.8	87.4	83.0	79.2	71.7	66.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.3	22.6	23.8	22.9	20.7	21.5	22.7	23.4	21.7	23.2	22.7	22.0		
		MCDBR	31.2	26.9	26.4	25.8	24.5	24.6	25.8	25.1	25.4	24.6	24.2	27.8		
		MCWBR	16.8	14.2	12.2	11.1	8.1	5.2	5.2	5.3	6.8	9.4	12.5	16.1		
	5% WB	MCDBR	23.0	20.0	21.0	19.8	18.3	19.5	20.9	22.3	18.7	18.6	18.1	21.7		
		MCWBR	15.6	13.8	11.4	10.5	7.5	5.3	5.1	5.2	5.7	9.2	12.0	16.5		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.322	0.349	0.389	0.424	0.417	0.428	0.419	0.396	0.366	0.343	0.315		
	taud		2.524	2.509	2.444	2.353	2.308	2.372	2.382	2.387	2.403	2.470	2.470	2.527		
	Ebn at Noon		292	300	297	287	276	276	272	274	278	280	279	286		
	Edh at Noon		28	30	35	40	42	39	38	38	36	31	29	26		
(44) All-Sky Solar Radiation	RadAvg		986	1165	1477	1767	1856	2138	2136	2016	1654	1354	1033	875		
	RadStd		107	182	136	164	191	152	174	111	156	198	129	116		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144

Nomenclature: See separate page